

Integrated Management Solutions for Durga Trader's Challenges

A Midterm report for the BDM capstone Project

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Contents

1 Executive Summary	03
2 Proof of Originality	04
3 Metadata	05
4 Descriptive Statistics	06
5 Detailed explanation of Analysis/Process	07
6 Results and Findings	08

Executive Summary

Durga Traders, a regional wholesaler of coal and related products, faces three critical operational challenges: volatile procurement prices, high reliance on credit-based sales, and inconsistent coal quality due to diversified sourcing. These issues have negatively impacted profit margins, cash flow stability, and customer retention.

To address these concerns, a structured data-driven approach was adopted. Sales and purchase data from May 2022 to April 2023 was collected, cleaned, and analyzed using tools such as pivot tables, VLOOKUP, and financial formulas. A profit margin analysis was conducted to compare monthly purchase and selling prices, while a cash flow analysis helped trace monthly revenue and expenditure patterns. Key descriptive statistics—such as mean, standard deviation, and range—were computed to understand the spread of costs and revenues.

Initial insights revealed that while the average revenue per transaction was relatively high, frequent bulk purchases and inconsistent pricing led to months with negative cash flow, such as July 2022 and April 2023. Credit-heavy sales in months like January and March 2023 further strained liquidity, even when profit margins appeared positive. These trends indicate that financial losses were driven more by irregular cash flows and credit risk than pricing alone.

This report aims to uncover the underlying causes behind these trends and suggest actionable strategies such as better supplier evaluation, credit control mechanisms, and dynamic pricing. The findings from the preliminary analysis will serve as the foundation for the final recommendations, enabling Durga Traders to make informed, data-backed decisions for long-term operational stability and profitability.

Proof of Originality

DURGA TRADERS

Ram Nagar Colony, Indra Nagar, Khair Road, Aligarh

Mobile No. - +91-8899920935

TO WHOMSOEVER IT MAY CONCERN

Subject- Giving consent to use data

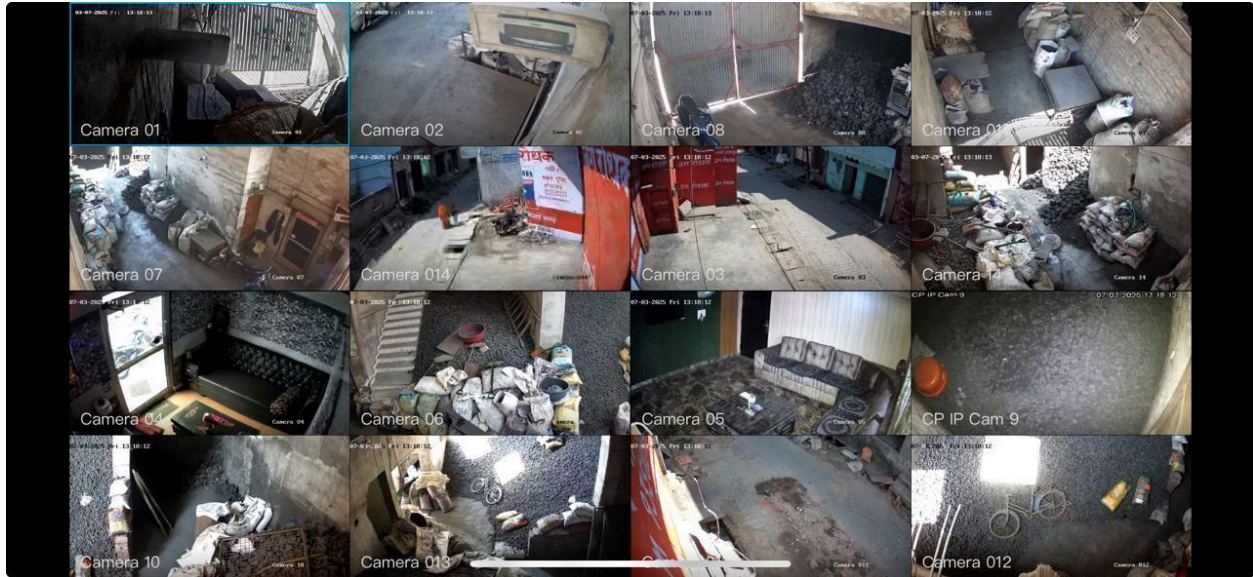
I, Porush Gupta, proprietor of Durga Traders give consent to Kratika Varshney to make use of the data that is provided for her project in the course Business Data Management of IIT Madras online degree program.



M/s Durga Traders
Ram Nagar Colony, Indra Nagar
Khair Road, Aligarh

Balance Sheet
1-Apr-22 to 31-Mar-23

Liabilities		Assets	
	as at 31-Mar-23		as at 31-Mar-23
Capital Account	28,75,891.25	Fixed Assets	21,64,898.04
Loans (Liability)	61,53,464.00	Investments	66,349.00
Current Liabilities	50,59,507.90	Current Assets	1,18,57,616.11
Suspense A/c			
Profit & Loss A/c			
Opening Balance			
Current Period	24,71,491.87		
Less: Transferred	24,71,491.87		
Total	1,40,88,863.15	Total	1,40,88,863.15



The camera view of their godown area.

3 Metadata

The dataset used in this analysis was sourced directly from Durga Traders' internal transaction records, covering the period from 01-April-2022 to 30-March-2023. The data was provided in Excel format and consisted of two primary sheets: Purchase Data and Sales Data, each structured in tabular format.

3.1 Data Sources

Purchase Data: Includes fields such as Date, Supplier, Quantity, Rate, and Particulars, which help track procurement patterns, cost fluctuations, and supplier reliability.

Sales Data: Contains Date, Particulars, Customer, Quantity, Rate, and Credit, allowing for an evaluation of revenue generation, customer purchasing trends, and outstanding credit dues.

3.3 Data Size

Purchase Data: 142 rows $\times$ 6 columns

Sales Data: 988 rows $\times$ 6 columns

3.4 Data Compilation and Processing

All data entries were cleaned and compiled into a master Excel file for analysis. New calculated fields, such as Profit per Transaction, Monthly Aggregates, and Credit Summary, were added using Excel formulas. The cleaned data was used to generate charts and statistical summaries across key metrics.

3.5 Data Access

The sheet containing all raw data, computed metrics, and visualizations will be submitted as a supporting file titled:

“  Durga_ Traders_Processed_Data ”

Please refer to individual sheets within the file:

Purchase_Data

Sales_Data

Descriptive_stats

Analysis

Charts

Descriptive Statistics

Outflow Transactions	Mean	78550.14774
	Median	11905
	Standard Deviation	191789.4767
Inflow Transactions	Mean	443155.6965
	Median	253800
	Standard Deviation	393490.2003
Financial Summary	Total expenses	77528995.82
	Total revenue	62210588.2

The mean inflow per transaction is significantly higher than the mean outflow per transaction, indicating that individual sales are of high value. However, the standard deviation for both inflow and outflow is large, suggesting high variability in transaction amounts—likely due to bulk orders and differing coal types or quality.

Despite high-value sales, the total expenses exceed total revenue, resulting in an operational loss over the recorded period. This imbalance, combined with observed transaction variability, points to the need for tighter control over procurement costs, optimized credit policies, and improved cash flow management.

Detailed explanation of analysis

Data collection and methodology

- **Data Sources:** Detail the ledgers and records provided by the business, specifying the period covered
- **Data Consolidation:** The process of extracting and compiling data into a master spreadsheet ensuring that the data was properly formatted and organised in columns and rows within the spreadsheet. This simplification made it easier to perform mathematical operations, such as calculating the profit margin.
- **Profit Margin Analysis:** The profit margin analysis involved comparing the purchase price and selling price of coal to calculate the monthly profit margin. Fluctuations in profit margins were linked to supplier costs and seasonal demand, helping to identify cost-saving opportunities.
- **Cash flow Analysis:** To understand overall sales and purchase trends, transactions were analyzed on a monthly basis. Revenue and expenses were compared to track cash flow, revealing months with higher sales volumes and those with lower activity.

Row Labels	Credit	Debit
2022-05	3066275.6	9040721
2022-06	2105521.09	8579632
2022-07	2532035.2	10685537.2
2022-08	1545511.3	1664760.6
2022-09	4576224.4	2063500
2022-10	6044716.2	1230909
2022-11	9521474.71	5496525
2022-12	7597552.56	3873290
2023-01	14503287.25	3120207
2023-02	13019439.3	5178019
2023-03	13012898.21	3712060
2023-04	4060	7839792.4
Grand Total	77528995.82	62484953.2

Fig 4.1: This is the monthly credit and debit data

Results and Findings

selling price and purchase price



Fig 5.1

Fig 5.1 shows the average monthly selling price (in blue) and the purchase price (in red) for the business. The purchase price varied significantly across months, with peaks in May 2022 (₹45,800/ton), while June 2022 (₹14,300/ton) had the lowest cost while selling prices remained relatively stable. This volatility disrupts pricing stability, making it difficult to maintain consistent profit margins. May 2022 had one of the worst margins, where the purchase price exceeded the selling price, leading to losses.

Some months saw a sharp increase in credit, such as January 2023 and March 2023, which could indicate bulk sales but also increased financial risk due to outstanding payments. July 2022 was a financial crisis point, with ₹2.5M revenue against ₹10.6M expenses, creating a severe cash flow deficit.

April 2023 is a red flag, with almost no revenue but very high expenses, indicating financial instability. Total credits exceeds debits, but the monthly volatility points to irregular cash flow management.

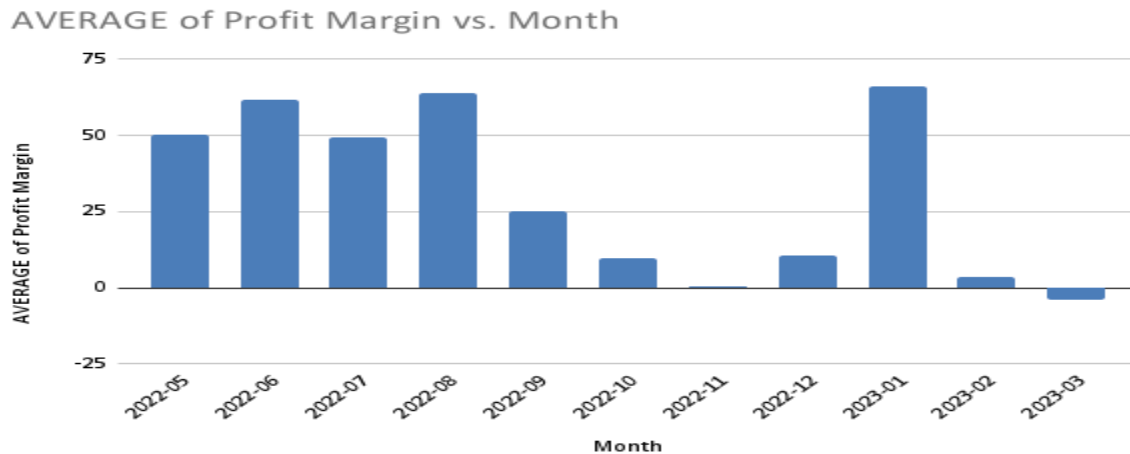


Fig 5.2

Fig 5.2 shows that the profit margin declined from Sep 2022 to end of the year, despite selling price being higher than the purchase price indicating that more sales were made on credit.

This indicates financial instability or difficulty in cash flow management.

March 2023 has negative profit margin irrespective of a higher selling price than purchase price likely due to clients may have deducted payments for substandard coal or they may have to sell the substandard coal at discount to clear excess stock.

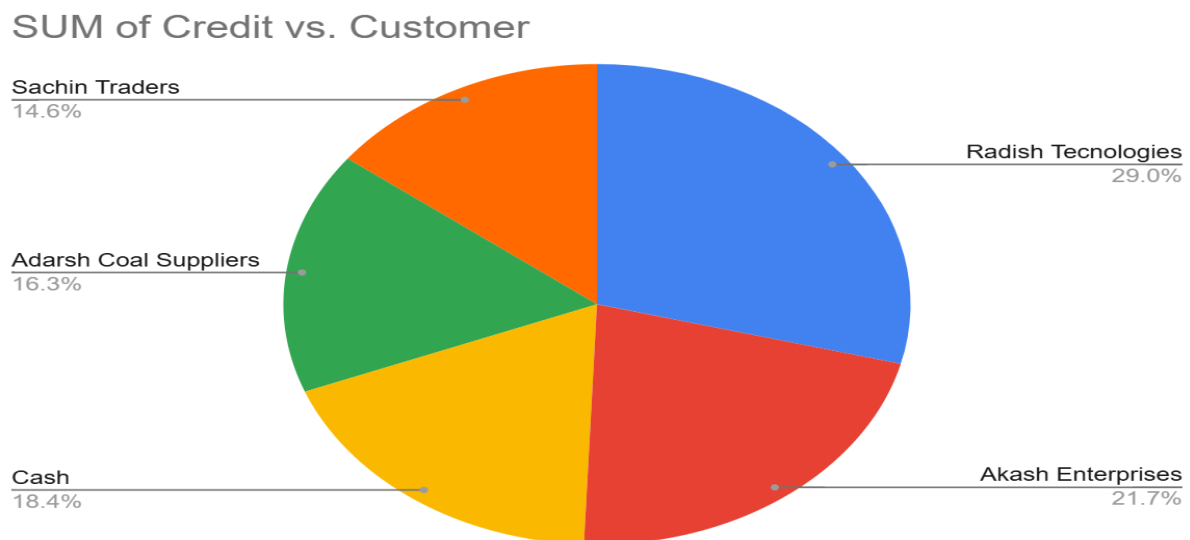


Fig 5.3

Fig 5.3 displays the distribution of credit sales among the top customers of Durga Traders. The data reveals that a significant portion of credit is concentrated among a few key clients:

- Radish Technologies holds the highest share (29%) of total outstanding credit, indicating a high dependency on their payments.
- Akash Enterprises follows with 21.7%, and Cash transactions surprisingly account for 18.4%, which may include delayed cash settlements.
- Adarsh Coal Suppliers (16.3%) and Sachin Traders (14.6%) round out the top five.

Summary and Next Steps

Through continuous data analysis and close collaboration with the business owner, the project is advancing effectively. Preliminary findings, as detailed in the attached reports, demonstrate the rigorous examination of sales, pricing, and financial patterns conducted to date. The analytical work remains ongoing, with a dedicated focus on extracting actionable insights that will inform strategic decisions, optimize operations, and foster sustainable growth for Durga Traders.

This partnership ensures that solutions are grounded in real-world data while aligning with the business's unique challenges and goals. Further updates will be shared as deeper insights emerge from the remaining analysis phases.